

# PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

## 1. EXECUTABLE ARTIFACTS AND LOGICAL DIRECTORY STRUCTURE

The complete functional deployment engine workspace includes the following standalone runtime configurations:

\* Directory Root/

```
|— app.py          [Core Flask Backend Micro-service Controller Router]
|— train.py       [Machine Learning Training Loop & Validation Setup Script]
|— requirements.txt [System Software Dependencies Environment Configuration]
|— data/
|   |— crop_data.csv [Historical Soil and Environmental Mapping Records Matrix]
|— models/
|   |— crop_model.pkl [Serialized Trained Scikit-learn Random Forest Binary Classifier]
|— templates/
|   |— index.html    [Responsive Bootstrap Layout Form and Alert Response UI View]
```

All internal modules communicate over real-time HTTP POST network protocols to stream floating-point environmental vectors directly into the active predictive binary.